



Mark Beyer Stjerne

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[Portfolio](#)

[GitHub](#)

[LinkedIn](#)

Astrophysicist specialising in optical data processing, from telescope operations to spectral analysis and pipeline development, bringing a multicultural perspective and a drive to contribute to real missions that help us understand our place in the universe.

EDUCATION

Master of Science in Physics (Astrophysics) (GPA: 10.0/12)

University of Copenhagen

Sep 2024–Present

Copenhagen, Denmark

Thesis title: *Mid- and far-infrared dust and gas properties of distant quiescent galaxies*

Relevant projects/coursework (☆ = Ongoing):

- ☆ **Master's thesis.** Developing Python pipelines to process and statistically analyse ALMA imaging and JWST spectroscopic data, examining the role of active galactic nuclei in driving star formation in distant quiescent galaxies.
- ☆ **Stellar Occultations of 3I/ATLAS.** Collaborating on an ongoing observational proposal to record spectra of the interstellar comet 3I/ATLAS during stellar occultations using the Nordic Optical Telescope in order to examine its chemical composition.
- ☆ **Niels Bohr Interferometric Teaching Telescopes (NBITT).** Assisting with the assembly and testing of a pair of interferometric teaching telescopes, including hands-on work on telescope hardware and tracking software development.
- Course: Advanced Methods in Applied Statistics.** Implemented Gaussian mixture models and two-dimensional KS testing on exoplanet population data to evaluate underlying distributional assumption; results presented in a written scientific report.
- Course: Astronomical Data Processing.** Wrote Python-based pipelines for raw imaging and spectroscopic data; performed data reduction, uncertainty propagation, and wavelength calibration to produce usable spectra and science images.

Bachelor of Science in Physics (GPA: 74.7/100)

University of New South Wales

Feb 2019–Dec 2021

Sydney, Australia

Relevant projects/coursework:

- Undergraduate Research Internship.** Analysed GALAH survey catalog data to examine lithium depletion in red-clump stars; applied data processing and visualisation in Python, presenting my results to a research group.

SKILLS & MORE

Languages/programs: Python, $\LaTeX$, Excel, PowerBI, Git, SQL

Python libraries: NumPy, Matplotlib, Pandas, Seaborn, SciPy, Astropy

Languages (CEFR): English (C2), Danish (C1), French (A2/B1), Japanese (A2)

Citizenship: Denmark, Australia

Hobbies/interests: Astronomy, languages/linguistics, ice skating, playing piano, photography

OTHER EXPERIENCE

Nordic Optical Telescope (NOT) Summer School 2025 [\[Link\]](#)

Aug 2025 (2 weeks)

Roque de los Muchachos Observatory

La Palma, Canary Islands

Selected to complete five nights of observing using the 2.56m NOT (ALFOSC imaging/spectroscopy and FIES echelle spectroscopy); carried out reduction and analysis of data, and presented findings at the school's closing mini-conference. Author on [GCN Circular 41483](#): 'GRB 250821A: NOT z-band upper limit' (2025) in reporting observations of a transient gamma-ray burst.

EMPLOYMENT

Claims Handler

DHL Insurance and Risk Management

Copenhagen, Denmark

Sep 2022–Present

Casual Teaching Assistant

University of New South Wales, School of Physics

Delivered Online

Feb–May 2022